

AstroStacker Pro

User Manual

Version 1.0.2

This manual covers the whole application: preparing and stacking your frames, the viewers, and all 36 processing tools.

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1. Before you start

AstroStacker Pro is a front-end. It does not reinvent stacking or processing - it drives established tools and adds a workflow, a viewer and a set of processing tools around them. The heavy lifting is done by Siril, with GraXpert and RC-Astro's Xterminator tools used where you have them installed.

What you need

- Siril is bundled - nothing to install.
- GraXpert is optional. Point the app at it in Advanced Settings if you want its background extraction and denoising.
- BlurXTerminator, StarXTerminator and NoiseXTerminator are optional and require your own RC-Astro licence.

Supported files

FITS (.fits, .fts, .fit), XISF, TIFF, PNG, JPEG, and camera RAW files. XISF is supported throughout - open, edit and save.

A note on stages. Some tools want LINEAR data - straight off the stack, before any stretch. Others only make sense on STRETCHED data. Where it matters, the tool says so in its own tooltip and often warns you if the image looks wrong for it. Chapter 9 walks through a full narrowband sequence in the right order.

2. Loading your frames

Frames go in through the left sidebar. Four buttons take your calibration and light frames, colour-coded to what they are:

Button	What it takes
Lights	Your actual exposures of the target
Darks	Same exposure and temperature as your lights, cap on
Flats	Even illumination, to correct dust and vignetting
Bias	Shortest possible exposure, cap on

Master calibration frames are recognised automatically. If you load a single master dark rather than a set of raw darks, the app detects it - by its name, or by the header keywords stacking software writes - and uses it directly instead of trying to stack it. Masters made in other software are recognised too.

Smart Import

The two Smart Import buttons read a folder, or a set of files, and sort them into lights, darks, flats and bias for you by reading their headers. Quicker than four separate trips through a file dialog, and it groups by exposure and temperature at the same time.

The file list

Everything you load appears in the Files Viewer tab, colour-coded by type, with exposure, temperature, filter and - once analysed - star count, FWHM and roundness. Tick and untick rows to control what gets stacked. Select several rows and press Enter, or double-click, to open them: the first goes to the main viewer and the rest into Mini Viewers.

3. Checking your frames

Three tools help you decide what to keep, and they answer different questions.

Analyze Frames

Measures every ticked light and fills in the star count, FWHM and roundness columns. Run this first - the other two build on it.

Subframe Selector

A dashboard of those measurements with charts for stars, FWHM and roundness. Sort and threshold to drop the frames that measure badly. Works before stacking, so you are not stacking frames you would have rejected anyway.

Blink Tool

Plays your ticked lights in sequence like a flip-book. This is the visual pass, and it catches what numbers do not: passing cloud, satellite and aeroplane trails, dew creeping in, a frame that drifted. Some bad frames measure perfectly well and are still obviously wrong once you see them move. Needs at least two lights ticked.

4. Stacking

Once your frames are loaded and checked, the settings above the START STACKING button control the run.

Key options

Option	What it does
Debayer	For raw one-shot-colour files. Leave OFF for mono or already-debayered data.
Drizzle 2x / 3x	Upscales during stacking. Useful for undersampled data; needs dithered subs.
Dark Optimization	Scales the master dark to better match your lights.
Equalize CFA	For colour flats, evens the Bayer channels.
Clean Hot & Cold Pixels	Uses the master dark to find and fix bad pixels.
Dual-Narrowband Extraction	Splits OSC dual-band subs into Ha and OIII - see below.
Auto-Open Preview	Opens the finished masters when the run completes.

Dual-band extraction

If you shoot a dual-band filter on a colour camera, this option splits each sub into separate Ha and OIII frames BEFORE registration and stacking. It works on the raw sensor data, before debayering, which matters: debayering interpolates, and that smears Ha from red photosites into the green and blue ones. Extracting first means each output pixel is genuinely measured light rather than interpolated.

Mixed exposures are fine. If you load 120-second and 60-second lights with a matching dark for each, they are calibrated separately with the correct dark and then merged into ONE master. Grouping only ever affects which dark calibrates what - it never changes how many masters come out.

5. The image viewer

The main viewer is where editing happens. Its toolbar:

Control	What it does
Open...	Loads any image. Select several and the rest open in Mini Viewers.
Undo / Redo	Steps back and forward through your edits.
Save	Save as a new file, or overwrite the original.
Mini Viewer...	Opens another viewer window.
Swap	Trades the main image with a Mini Viewer's.
Flip H / Flip V / 90	Orientation.
Fit / 100%	Zoom to fit, or to actual pixels.
STF Stretch	A display-only auto-stretch. Does NOT change your data.
Linked / Unlinked	Whether the display stretch treats channels together or separately.

STF Stretch only affects what you SEE. Linear data looks black without it, which is why it defaults on. Turn it off to view the real values - and remember that the stars-only image from StarXTerminator opens with it off deliberately, because an autostretch on a sparse star field hauls the background noise up and looks wrong.

You can also drag image files straight onto the viewer. Drop several and the first opens in the viewer with the rest in Mini Viewers.

6. Mini Viewers

Mini Viewers are additional windows for looking at other images while you work - comparing a before and after, keeping your Ha next to your OIII, or holding a result you are not ready to commit to.

- Open... loads an image; you can also drag one in.
- Save As... writes the image to a file, from disk rather than from the screen, so full bit depth is kept and the display stretch is never baked in.
- Swap trades this image with the main viewer, so you can edit it with the full tool set and swap back afterwards.
- Minimize tucks the window into a small icon in the main window rather than the taskbar.

Several things arrive in a Mini Viewer UNSAVED and live only in the temp folder until you save them: the stars-only image after StarXTerminator, Split Channels output, and Pixel Math results. Use Save As... to keep them.

7. How editing works

Tools fall into two groups, and the difference is worth understanding.

Tools that commit straight away

Simple Stretch, GHS, Curves, VeraLux, Camera RAW, the colour tools, Pixel Math, Bin/Downscale, Flip and Rotate apply directly to the working image with no save prompt. That is deliberate: these are tools you tweak repeatedly, and a prompt after every adjustment would be maddening. Build up your stretch across several

small steps and use Undo freely.

Tools that ask

Crop, Background Extraction, GraXpert, SPCC, the Xterminator tools, Star Reduction, Screen Stars, Blemish Blaster, Dust Lane Enhancer, Dark Structure Enhance and Narrowband Normalization offer Save, Overwrite or Keep Editing when they finish. These are single decisions rather than iterative adjustments.

Tools that build something new

Combine LRGB, NB to RGB Stars and RGB Stars to NB take several images and make another. Their sources are strictly read-only. The result goes to the main viewer and whatever was there moves into its own Mini Viewer, so nothing is lost and no source master can be overwritten.

The Overwrite button in the save prompt names the file it would replace, so you always know what is about to change. Save As New is always the safe choice.

8. The processing tools

All 36 tools, in the groups they appear in on the right-hand sidebar. Each description is the tool's own in-app help, so this section and the app can never disagree with each other.

Masking

RangeSelection Mask

Generates a mask from a range of pixel values, per PixInsight's own RangeSelection tool - Lower/Upper limit, Fuzziness (soft transitions), Smoothness (blur), Screening (proportional rather than flat white), Lightness (CIE L* vs per-channel RGB/K), Invert. Check 'Active' to apply it to the live image: white areas get fully processed by whatever tool runs next, black areas stay completely untouched - works with any tool in the app, not just this one. This window must stay open for the mask to stay active.

Image Prep

View FITS Header

Shows the raw FITS header (exposure, temperature, filter, WCS, etc.) for the file currently being previewed.

Bin/Downscale

Shrinks the image by an exact integer factor (true pixel binning - averages each block down to one pixel, not a blurry resize) or a custom percentage. Useful for making a smaller version of a finished image to share online. Applies directly, same as GHS/Curves - use Undo to revert.

Set Preview Region

Optional: mark a region to keep at TRUE full resolution while adjusting sliders in GHS or Curves, while the rest of the image stays on the fast downscaled preview - like Siril's own ROI feature. Off by default and doesn't replace the normal preview.

Crop Tool

Draw a crop box on the image, then drag its edges to adjust, and click Apply Crop when ready. If any Mini Viewer windows are open, you'll be offered the same crop for those too.

Blemish Blaster

Franklin Marek's Blemish Blaster technique - drag on the image to mark a spot (hot pixel, dust donut, satellite trail fragment) for repair. Repairs by sampling 3 nearby regions that brightness-match the spot's own surroundings, avoiding other stars or gaps rather than just blurring the neighbourhood. Feather and Opacity sliders control how strong and how soft the repair is.

Calibration & Astrometry

Image Solver

Runs just Siril's plate solve on its own, embedding WCS astrometry (RA/Dec, pixel scale, rotation) into the file's header - no colour calibration afterward, unlike SPCC below. Writes straight back to the same file, since only header metadata changes. Requires an internet connection - run this after cropping to avoid losing the plate solve to the crop.

SPCC

Runs Siril's plate solve + Spectrophotometric Color Calibration on the currently previewed image. Requires an internet connection. Run this after cropping to avoid losing the plate solve to the crop.

LinearFit

Fits a robust straight line matching the currently-open image's pixel values to a separate reference image's, per channel, then applies that line to the whole image. Useful for matching brightness/background levels across separately-processed masters (e.g. different narrowband filters) before combining them. Images must be pixel-for-pixel registered.

Background Tools

GraXpert

Runs GraXpert's AI Background Extraction or Denoising on the currently previewed image. Requires the GraXpert path to be set in Advanced Settings -> AI Tools.

Background Extraction

Runs Siril's background extraction (subsky -rbf) on the currently previewed image. Useful for applying it AFTER other processing like NoiseXTerminator instead of only automatically before stacking, since running noise reduction on already-extracted data can give patchy results.

Background Neutralization

Equalizes the average red, green and blue background levels to a neutral grey, removing any colour cast in the sky background. A neutral background is a precondition for correct colour calibration, so this normally runs before SPCC or other colour calibration steps. Draw a Preview Region over open sky first to sample just that area, or leave it unset to use the whole image.

Noise & Sharpening

Xterminator Tools

Opens BlurXTerminator, StarXTerminator, and NoiseXTerminator in one tabbed window, each running on the currently previewed image via the RC-Astro CLI. Requires the CLI path to be set in Advanced Settings -> AI Tools.

Star Tools

RGB Stars to NB

Puts REAL star colour, measured from BROADBAND data, onto a narrowband image - the complement of NB to RGB Stars, which synthesises star colour when you have no broadband data at all. Needs three registered, stretched images: your RGB frame with stars (an unfiltered OSC frame counts, but NOT one shot through a narrowband filter - its star colour is already false), the same RGB frame starless, and your narrowband image starless. It recovers the star layer from the RGB by unscreening, screens it onto the narrowband, then does a CIELAB transfer taking lightness from one and colour from the other, which is what makes it more than a screen blend. Everywhere there are no stars the narrowband image is preserved exactly. Ported from Bill Blanshan's RGB Stars-to-Narrowband Transfer V14.

Star Reduction

Bill Blanshan's Star Reduction PixelMath methods (Transfer/Halo/Star) - needs a starless version of your image (e.g. from StarXTerminator) alongside the original. Select both images to begin.

Screen Stars

Recombines a starless image with a separately-processed stars-only image via the Screen blend - Bill Blanshan & Mike Cranfield's technique, the exact inverse of the "unscreen" math used to remove stars in the first place. Select both images to begin.

Star Stretch

Franklin Marek's Star Stretch v2.1 technique - a linear-to-nonlinear stretch tuned for a stars-only image, boosting colour and removing green cast automatically. Meant to run on a combined stars image before screening it back into a starless one.

NB to RGB Stars

Franklin Marek's NBtoRGBStars v1.6 technique - combines Ha/OIII/(optional SII) stars-only images into a realistic RGB stars image, ready to screen back in. Also accepts a single OSC image shot through a dual-band filter instead of 3 separate mono ones. Optional Apply Star Stretch runs that technique on the result as the final step.

Stretch Tools

Simple Stretch

Permanently applies the same auto-stretch algorithm used for the viewer's own display to the actual pixel data - a quick one-slider stretch. Needed before tools that expect a stretched image (e.g. Narrowband Normalization) if you haven't stretched elsewhere.

GHS

Built from scratch in this app (not scripted through Siril) from the published GHS specification (David Payne & Mike Cranfield) - Generalized Hyperbolic, Midtone Transfer, Arcsinh, Power Law, or Linear, with full control over strength, symmetry, and shadow/highlight protection, plus Colour/Lightness/Saturation modes. Live preview.

Curves

Interactive curve editor - Full RGB, Individual RGB, Luminance Only, or Saturation. Click to add a point, drag to move it, right-click to remove it. Built directly in this app (not scripted through Siril).

VeraLux HyperMetric Stretch

A ported, verified version of Riccardo Paterniti's real VeraLux HyperMetric Stretch script for Siril - a linear-to-nonlinear stretch that preserves each pixel's original colour ratios rather than stretching R/G/B independently, avoiding the hue shift a plain histogram stretch can cause. Ready-to-Use (finished, export-ready look) or Scientific (raw, deterministic - a base for further manual tone mapping) modes, with sensor-specific luminance weighting and an Auto-Calculate solver.

Tone

Camera RAW Editor

AstroStacker Pro's own expanded version of the original Brightness/Contrast/Sharpen tool - adds Exposure, Highlights/Shadows/Whites/Blacks, Vibrance, Clarity/Texture, and an adjustable-radius, noise-aware Sharpen, alongside the original 4 controls. Live preview. Only really works on already-stretched (non-linear) data - on linear data, where almost all the signal sits near black, most of these controls behave unpredictably or barely at all.

Structure & Detail

Dust Lane Enhancer

Detects dust lanes and dark filaments by their SHAPE - the Frangi vesselness algorithm from medical blood-vessel detection (dust lanes are the same mathematical structure: elongated ridges) - then deepens only what it detects. Not a brightness stretch: a shape-based enhancement no other astro processing software has. Works best on a starless image - bright stars can produce a faint ring-like false response of their own. Use Show Detection Map to see exactly what it's found while tuning.

Dark Structure Enhance

Carlos Sonnenstein & Oriol Lehmkuhl's DarkStructureEnhance (the real PixInsight script, ported faithfully) - builds a mask of dark large-scale structure via wavelet layer decomposition, then darkens through it, deepening dark nebulosity and dust against its surroundings while stars and bright structure stay untouched. Companion to the Dust Lane Enhancer: this boosts dark structure generally, whatever its shape, vs that tool's filament-specific detection. Layers controls what counts as large-scale; Show Mask displays exactly what's selected.

Multiscale Local Contrast

Boosts contrast only at the structure SIZES you choose, leaving every other scale mathematically untouched - a genuinely different lever from a stretch or curve, which move every scale together. Splits the image into wavelet scale layers (the same verified transform Dark Structure Enhance uses), amplifies just the selected band, and rebuilds it exactly. Background Protection holds the effect back over faint sky, where boosting would otherwise amplify noise harder than real structure - measured directly on a real image, which is exactly why it defaults on. Note it boosts STARS as well, since a star is small high-contrast structure by definition (measured: bright star bodies gained +0.105 on a real frame at default settings) - best run on a starless image, or with Strength kept moderate.

Colour Tools

Astro Color Mixer

Hue-targeted colour and luminance refinement, adjusting each colour band (Red/Orange/Yellow/Green/Cyan/Blue/Purple/Magenta) independently, with built-in protection for shadows, near-neutral pixels, and highlights/stars. Adapted from Pat Cosgrove's original PixInsight Astro Color Mixer script (Cosgrove's Cosmos), via Yannick Dutertre's (Cuiv the Lazy Geek) Siril port. Only really works on already-stretched (non-linear) RGB data.

Color Masks

Bill Blanshan's hue-based colour masks - isolates one colour family (Red/Yellow/Green/Cyan/Blue/Magenta) with adjustable strength and blur, so a saturation adjustment only affects that colour, leaving everything else untouched. Invert Mask is on by default.

Color Saturation

A saturation multiplier per hue (0=unchanged, positive boosts, negative cuts) at each of the 6 fixed hue positions, smoothly interpolated between them, plus a Hue Shift to rotate the whole set together. Uses fixed control points at Color Masks' same 6 hue positions rather than a fully free-form curve.

SCNR

Subtractive Chromatic Noise Reduction - corrects a colour cast from one channel (usually green) being misaligned with the other two, protecting star colour in the process. All 5 protection methods (Average/Maximum/Minimum Neutral, Maximum/Additive Mask), plus Preserve Lightness.

Correct Magenta Stars

Removes the magenta fringing chromatic aberration leaves around stars - usually worst toward the frame corners, where the optics focus red and blue at slightly different points from green so the star's edge shows the colour green is missing from. It lifts that deficient green back toward its neighbours, which is the OPPOSITE clamp to SCNR: SCNR pulls an excess channel DOWN, this fills a deficient one UP. By default it only touches stars, found with the same detector the palette tools use and grown outward to cover the fringe ring rather than just the core - so magenta nebulosity, which is real signal in narrowband and palette images, is left alone. Turn that off for a broadband image with an overall magenta cast.

Luminance-preserving, so removing a fringe does not brighten the star's edge.

Channels & Palettes

Split Channels

Splits the open image into separate MONO images, one per channel, each opened in a new Mini Viewer - the source is read-only and never altered. Not needed for palette remapping (PixelMath does any mapping in one expression), but it is the only way to process each channel SEPARATELY: stretch a weak OIII harder than the Ha, denoise only the noisy one, remove stars from one and not the other, then recombine with Combine LRGB or NB to RGB Stars - both of which expect separate mono masters that a one-shot-colour frame could not previously produce inside the app. On a duo-band OSC frame, Red is your Ha and Green/Blue carry the OIII. Optionally also extracts a synthetic Luminance.

PixelMath

Applies an arithmetic expression to every pixel - one expression for all three channels, or one per channel to mix or swap them. Symbols R, G, B, L (luminance), T (whole image); functions min, max, abs, sqrt, log, exp, pow, mtf, iif; and statistics med, mean, sdev, mad that collapse an image to a single number, so things like 'R - med(R) + 0.5' work as they do in PixInsight. Live preview in the main viewer as you type. Expressions are parsed and whitelisted rather than executed directly, so a typo reports a plain error instead of doing anything unexpected.

Combine LRGB / Narrowband

Combines separate mono master images into one colour image - useful after stacking a mono camera or extracting individual narrowband filters. Custom mode assigns any three images to Red/Green/Blue; HOO/SHO/HSO/HOS presets map your narrowband filters automatically. An optional Luminance master can be blended in for LRGB/LSHO-style results.

Narrowband Normalization

Bill Blanshan and Mike Cranfield's Narrowband Normalization process (HOO/SHO/HSO/HOS) on the currently previewed image. Best run on a starless, background-neutralized image.

Dual-Band to SHO

Turns a stacked dual-band / OSC colour image into a gold-and-teal SHO-style LOOK. NOT the same job as Dual-Band Extraction on the stacking side, despite the similar name - that one splits your raw CFA subs into genuine Ha and OIII masters before stacking, from real measured photosites. This works on an already-stacked colour image and SYNTHESISES the look: a duo-band frame contains no sulphur at all, so this CREATES a palette rather than revealing one - which is exactly why it is labelled as such. What IS real in the data is the OIII/Ha ratio, and this amplifies how each pixel's ratio DEVIATES from the field median, since the measured spread is small enough that merely scaling channel levels only tints the frame. Brightness is never invented - every pixel keeps the exact luminance it started with, verified to 1.8e-07 - so it changes colour only. Original code, verified on three different real frames. Chains with Narrowband Normalization rather than competing: that tool BALANCES a palette already present, this one creates one.

9. Working with narrowband

Dual-band data has more routes through the app than anything else, and the order matters. This is a full sequence with the reasoning.

Shooting a dual-band filter on a colour camera

1. Load your lights and calibration frames, and tick Extract Dual-Narrowband Filter before stacking. You get separate Ha and OIII masters, extracted from the raw sensor data before debayering.
2. If you did not extract during stacking and have a stacked colour master instead, use Split Channels. In duo-band mode it gives you Ha and OIII as separate mono files. Choose the method:
 - Simple takes the channels as they are - Ha from red, OIII from green or green and blue. Works at any stage.
 - Sensor Un-Mix inverts your sensor's crosstalk instead, so each output holds far less of the other line. LINEAR DATA ONLY - use it straight off the stack.
3. Process each channel separately. This is the real value of splitting: stretch a weak OIII harder than the Ha, denoise only the noisy one, remove stars from one and not the other.
4. Recombine with Combine LRGB / Narrowband. Pick a palette:

Mode	What it does
Custom (RGB)	Assign any three mono images to Red, Green and Blue
HOO (Bicolor)	Ha to red, OIII to green and blue - the natural two-channel choice
SHO (Hubble)	SII to red, Ha to green, OIII to blue
HSO / HOS	Other fixed orderings
Foraxx (adaptive)	Blends per pixel from the data itself - STRETCHED data only

5. Balance with Narrowband Normalization, which brings the channels into a sensible relationship, and use SCNR if a green cast remains.

The shortcut

If you only want the SHO look from a dual-band image and do not need the channels separately, Dual-Band to SHO does it in one step from the stacked colour image. It SYNTHESISES the palette rather than extracting real channels - the SII is derived, not measured - and it wants stretched data.

Two tools, opposite requirements, easy to mix up. Sensor Un-Mix is LINEAR only: it subtracts a background and divides by sensor coefficients, which assumes values still proportional to captured light. Foraxx is STRETCHED only: its per-pixel blend needs values spanning the full range, and on linear data it collapses to a plain mapping. Both warn you if the data looks wrong for them.

Star colour

Narrowband stars are a known problem - they come out magenta or simply wrong, because the filter has thrown away most of the colour information. Two tools address it from opposite directions. NB to RGB Stars SYNTHESISES plausible star colour from the narrowband data itself, for when that is all you have. RGB Stars to NB transfers REAL colour measured from broadband frames, which needs an unfiltered OSC or RGB exposure of the same target, registered to the narrowband image.

10. Masks

RangeSelection Mask builds a mask from the image's own brightness, and once a mask is active, tools apply through it - so an adjustment can be confined to the bright parts, the faint parts, or anywhere in between.

The usual uses are protecting stars from a stretch that would bloat them, confining noise reduction to the background where the noise actually is, and keeping a contrast boost off the core of a bright object.

Color Masks works on hue instead of brightness, isolating a colour family so an adjustment only touches, say, the red regions.

11. Troubleshooting

Symptom	What it usually means
Stacking stops with 'No sequence found'	A calibration group held a single frame. The app now uses a lone frame directly as its master, so this should not recur.
A tool looks like it did nothing	Check whether it needs linear or stretched data. Several tools warn in their status line when the image looks wrong for them.
SPCC says the solution is imprecise	Usually the data rather than a fault. Narrowband filters throw away most of the star colour SPCC calibrates from, so the fit is weak. The console prints a summary with the fit quality after every run.
Colours look wrong after SPCC	SPCC calibrates from STARS. On reflection nebulae, getting the stars right does not guarantee the dust renders as your eye expects. Judging by eye is defensible there.
The image looks black	Linear data. Turn on STF Stretch to view it - that is display-only and does not change the data.
Magenta fringes around stars	Chromatic aberration, worst toward the corners. Correct Magenta Stars fills the deficient green back in.
An overall green cast	SCNR, which is the opposite operation - it pulls an excess channel down.

The console

The console along the bottom logs everything the app and its underlying tools do. Copy Log puts the whole thing on the clipboard and tells you how many lines went; Clear Log empties it. When testing something, clear first and then copy, so you get just that run.

12. Credits and licence

This app is a front-end. Full credit for the actual stacking and processing work goes to the people and projects below.

Whose work	What it provides
Siril	The stacking, calibration, registration and SPCC engine
GraXpert	Background extraction and denoising
RC-Astro	BlurXTerminator, StarXTerminator, NoiseXTerminator
Bill Blanshan & Mike Cranfield	Narrowband Normalization, Screen Stars, SCNR method

Bill Blanshan	Star Reduction and Color Masks methods; RGB Stars to NB
Dave Payne & Mike Cranfield	Generalized Hyperbolic Stretch
Franklin Marek / Seti Astro	Star Stretch, NBtoRGB Stars, Blemish Blaster
Riccardo Paterniti	VeraLux HyperMetric Stretch; the un-mix port
Raul Hussein / Astrocitas	DBXtract, the origin of the sensor un-mix method
PixInsight / Pleiades Astrophoto	RangeSelection reference

AstroStacker Pro is released under the GNU General Public License v3. The source is included with every release.